a Reference membership

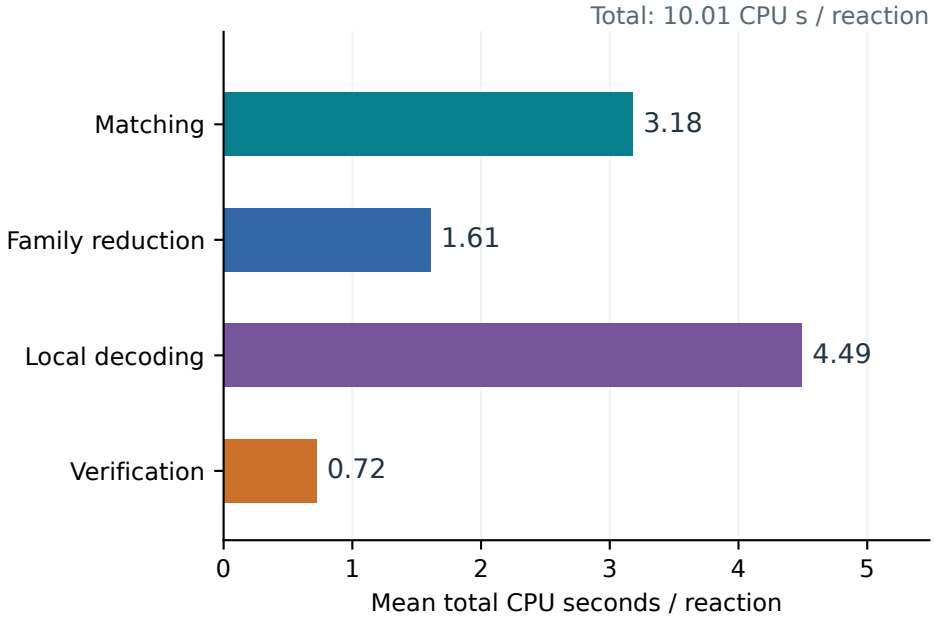
98.87%

1,830 of 1,851 reactions

21 not recovered

Both directions evaluated
No unresolved outcomes

b Recorded stage costs



Completed Golden validation · both directions included · CPU cost, not elapsed time